

4 SPAN STABILITY ASSESSMENT

4.1 Summary

This chapter outlines the assessment procedure. It has been developed out of work originally conducted in ACARP Project C5007 and subsequently enhanced within the current project, C8033. Tools for conducting the assessment, including the analytical tools, numerical tools and empirical tools are briefly described. Examples of applying these tools to the span stability assessment are demonstrated.

4.2 Introduction

The recommended procedure for unsupported span stability assessment involves three main steps:

Identify the span failure mechanisms;

Select proper tools for span stability assessment;

Assess the stability of entry spans using all of the available geological and geotechnical data.

Each of the three steps is discussed in details below.

4.3 Unsupported failure mechanism.

Unsupported span failure mechanisms are controlled by a number of factors including rock strength, in situ stresses, joints, lithology, sedimentology, and excavation geometry. In general, failure mechanisms can be divided into five main types. They are:

Span delamination and snap-through (Figure 4.1). Span failure starts with delamination, followed by tensile cracking of the immediate roof layer, which finally breaks due to snap-through. This type of failure often occurs in roofs with weak inter-lamination strength and low to moderate in situ stresses.

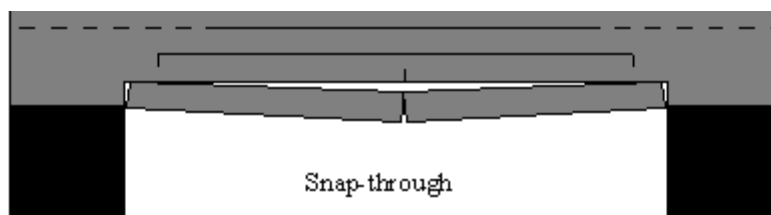


Figure 4.1. Span instability caused by roof delamination and snap-through.

Coal beam shear failure (Figure 4.2). When the immediate roof is a layer of cleated coal, it may fall from shear failure along existing vertical cleats. Lack of horizontal confinement and loading from overburden rock mass contribute to this type of failure.

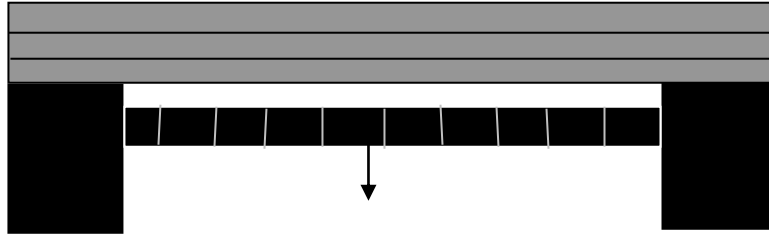


Figure 4.2. Coal roof instability caused by shear failure along coal cleats.

Block fall (Figure 4.3). This type of span failure often occurs in strong roof rocks where joints are well developed. In this case, joint distribution and joint strength dictate span stability.

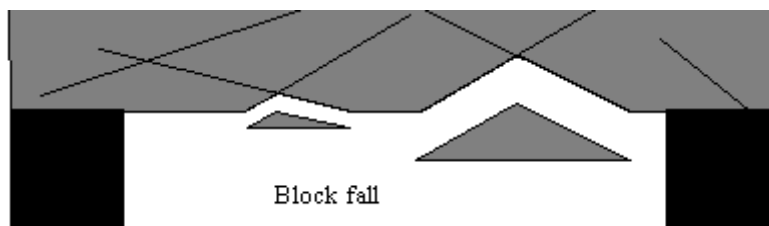


Figure 4.3. Span instability caused by block falls.

Intact rock shear failure (Figure 4.4). This mechanism often occurs where the roof rock is weak but relatively intact and the in situ stresses are high in comparison to the rock strength. Spans may fail due to this mechanism in underground roadways and, sometimes, highwall mining entries in a high horizontal and vertical stress regime.

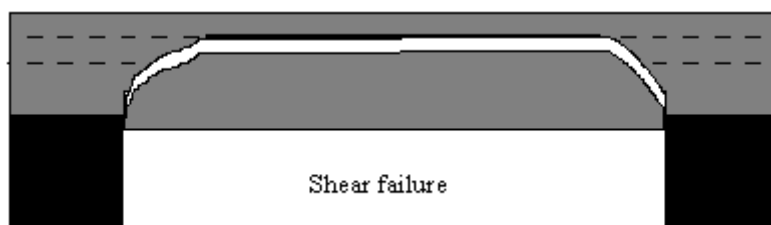


Figure 4.4. Span instability caused by shear failure.

Span buckling (Figure 4.5). Buckling failure occurs when the horizontal stress is relatively high, the roof strata are thinly laminated and the inter-lamination strength is low.

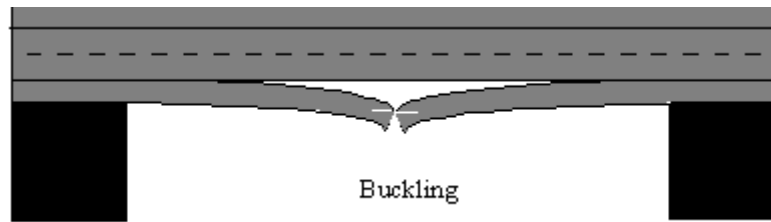


Figure 4.5. Span instability caused by buckling.

For typical Australian highwall mining conditions (laminated roof with low to moderate in situ stresses), the most likely failure mechanisms are the snap-through and coal beam shear mechanisms. Joint controlled block falls are also common in areas where intensive jointing has occurred in the immediate roof.

Two analytical models have been developed within the ACARP C5007 and C8033 projects for the assessment of span instability due to the snap-through and coal beam shear failure mechanisms. Block fall failure mechanisms should be analysed using numerical tools such as UDEC (Itasca, 1996) and 3DEC (Itasca, 1994). The shear failure mechanism and the bulking failure mechanism are not very common in highwall mining conditions. They may be assessed by using existing numerical codes for continuous or discontinuous media.

4.4 Span stability assessment tools

4.4.1 The Laminated Span Failure Model (LSFM)

As part of a project partially funded by ACARP (C5007, Geomechanical Factors Affecting Highwall Mining), the LSFM was developed by CSIRO specifically for highwall mining span stability assessment in the thinly laminated roof strata that often occur at highwall mining sites in the Bowen Basin of Queensland. The LSFM models the roof as a series of thin rock beams which are detached from each other and which can progressively fall out of the roof. The LSFM combines an equation to calculate the maximum stable span with a simple technique to estimate the likelihood of span failure and the height of fall for a given highwall mining span.

In two dimensions, the LSFM considers the immediate roof layer as a thin and long beam which is detached from the overlying rock. Cracks can develop near the abutments and at the mid-span as the beam deflects under its self-weight. The final failure of the immediate roof beam is caused by snap-through at the mid-span, see Figure 4.1. According to the LSFM, the maximum stable span is controlled by the roof layer thickness, Young's modulus and compressive strength, and load applied to the layer (Shen and Duncan Fama, 1996).

The maximum stable span is easily calculated by using a simple closed-form solution as follows:

$$S = 1.873 \cdot t \cdot \left(\frac{E}{p} \right)^{1/4} \cdot \delta \left(\frac{\sqrt{Ep}}{\sigma_c} \right) \quad (4.1)$$

where

S — span (m)

t — roof layer thickness (m)

E — Young's modulus (MPa)

P — distributed vertical load on the roof layer (MPa), in most cases being the self-weight of the layer. But it can be greater due to additional load from the overlying layers, particular when the overlying layers are not horizontal.

δ — a dimensionless factor to consider rock crushing (≤ 1.0). It is a function of the compressive strength, Young's modulus of the roof rock and the load.

σ_c — Unconfined compressive strength (MPa)

To take account of the likely variability in roof conditions across a highwall mining reserve, the LSFM incorporates a simple technique to assess the probability of span instability. This is required because a highwall mining reserve (HMR) typically extends over an area of 0.3 to 0.5 km². roof conditions will vary over an area of this magnitude. One way to account for this variation is to define a range of possible values for each parameter and assign a probability to the high, low and most likely value. For instance, if we know that the Young's modulus of the roof rock is likely to be in the range 3-12 GPa and considered most likely to be 8 GPa, we may define three values for this parameter, see Table 4.1.

Table 4.1. Young's modulus values and their assigned probabilities.

Low Value		Most Likely Value		High Value	
E_1	$p(E_1)$	E_2	$p(E_2)$	E_3	$P(E_3)$
5 GPa	0.2	8 GPa	0.6	11 GPa	0.2

The probability, $p(E_i)$, $i = 1,2,3$, assigned to each value, represents a subjective judgment as to its likelihood of occurring based on the best available information. This can be adjusted and the analysis re-run if and when new information becomes available.

All the input parameters used in Equation (4.1) can be defined in this way. Using basic statistical theory, the assessment results can be presented in a probabilistic way. Typically results might be shown as a graph of span width versus the cumulative probability of span failure (Shen and Duncan Fama, 1996).

The height of possible roof falls for a given span is also estimated in the LSFM. It is assumed that the failure tends to create an elliptical opening whose shape is determined by

the ratio of the horizontal and vertical stresses. Details of the calculation are described in Shen and Duncan Fama (1996). An example case is given in Section 4.5.

4.4.2 The Coal Roof Failure Model (CRFM)

In many cases, a coal layer is left in the immediate roof either because of a limitation in the mining height or with the aim of improving span stability. When roof coal is left, it is the stability of the coal that is the issue and not necessarily the stability of the non-coal strata. The mechanism of coal falls is notably different from that of laminated roof rocks, due to the structural differences between the two materials. Rock falls often occur as thin slabs, but roof coal failures often incorporate the whole thickness of the unmined coal. The dominant mechanism in coal roof failures is shear failure along sub-vertical cleats (Figure 4.2), which is often associated with low horizontal stresses in the coal seam.

A typical failure mechanism of roof coal is shown in Figure 4.6. After the excavation of an entry, the roof-coal beam deflects in response to the vertical stress from the main roof, while the main roof deforms in response to stress redistribution in the vicinity of the entry. The coal beam often deflects in a different manner from the main roof, and when it does, the coal beam separates maximally from the main roof in the central part of the span. However, contact between the coal beam and the main roof still exists near the ends of the span. There, the coal beam is loaded mainly vertically, through the contact with the main roof. The vertical load on the beam leads to high shear stress in the beam cross-section at the ends of the span. If the resultant shear stress is high enough and the horizontal confinement is low enough, shear failure will occur and the whole coal beam will fall into the entry.

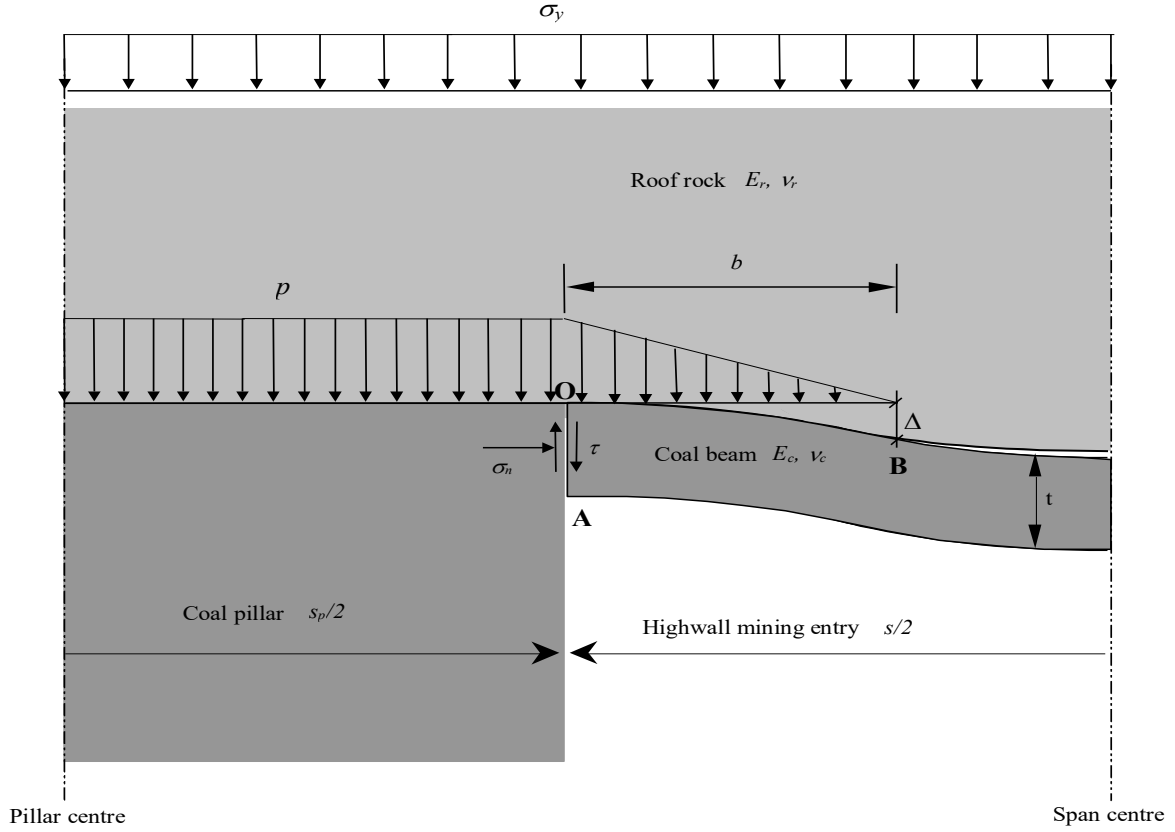


Figure 4.6. A coal roof failure model.

A formula for this type of failure has been derived. The details are given in Shen and Duncan Fama (1999). In summary, the Factor of Safety of the coal beam is given by the equation below:

$$FoS = \frac{\left(\sigma_x + \frac{b(s + s_p)\sigma_y \tan(\phi_i)}{2t(b + s_p)} \frac{(s - 2b/3)}{(s + s_p)} \right) \tan \phi + c}{\frac{1}{2} \frac{b}{t} \frac{s + s_p}{b + s_p} \sigma_y} \quad (4.2)$$

where

- ϕ, c — friction angle and cohesion of sub-vertical coal cleats;
- σ_x — in situ horizontal stress in coal seam;
- σ_y — overburden stress;
- ϕ_i — friction angle of coal/roof interface;
- s, s_p — entry width and pillar width;
- t — thickness of roof coal;

b — contact length between the coal beam and roof rock.

The contact length b is the solution of the following equation:

$$\frac{2(1-\nu_r^2)}{\pi} \frac{b}{t} \left[\frac{3}{2} - \ln b + \ln s_p \right] - \frac{E_r}{E_c} \left[\frac{b^4}{10t^4} \left(4 - 5 \frac{b}{s} \right) + \frac{(1+\nu_c)}{3} \frac{b^2}{t^2} - 1 \right] = 0 \quad (4.3)$$

where E_r/E_c is the stiffness ratio of the roof rock (E_r) and coal (E_c);
 ν_r and ν_c are Poisson's ratio of the roof rock and coal.

An example of the assessment of coal roof stability is given in Section 4.5.

4.4.3 Numerical tools

If the span failure is caused by a combination of two or more different mechanisms or the geology is too complicated to use the analytical model, then numerical tools may be used. The numerical methods used to analyse roof stability can be classified into two categories: the continuum approach and the discontinuum approach.

In the continuum approach, roof rocks are modelled as elasto-plastic materials. Intact rock yielding is considered to be the indicator of roof failure. There are many finite element and finite difference codes which use the continuum approach. The most commonly used codes in stability assessment of underground openings are FLAC and its three-dimensional version FLAC^{3D} (Itasca, 1995, 1996).

When using FLAC to predict the stability of entries, the rock mass is treated as a continuous medium. Bedding planes can be simulated using the ubiquitous joint model, which considers possible failure along the bedding planes without involving intact rock. The advantage of using FLAC for span stability prediction is the simplicity of the numerical model and the calculation speed. However, FLAC modelling does not predict roof falls explicitly but only provides an indication of the extent of roof plasticity. In an actual mining span, plastic yield in the roof does not necessarily lead to roof falls because the yielded roof rocks can still be held in place by horizontal confinement and friction.

Recently CSIRO has developed a Cosserat continuum code for span stability assessment. The Cosserat model (Cosserat and Cosserat, 1909) provides a large-scale (average) description of a layered medium. In this model, inter-layer interfaces (joints) are considered to be smeared across the mass, i.e. the effects of joints are implicit in the choice of stress-strain model formulation. An important feature of the Cosserat model is that it incorporates bending rigidity of individual layers in its formulation and this makes it different from other conventional implicit models such as the ubiquitous joint model (Itasca, 1995) which is based simply on strength anisotropy theory. Thus the code considers the presence of bedding planes in the roof by including the effect of such bedding planes in its mathematical formulation. The code may also model rock creep and therefore is able to simulate time-

dependent span stability. Application of the Cosserat code in underground roadway span assessment has shown promising results (Kelly et al., 2001).

In the discontinuum approach, the roof strata are considered to be an assembly of blocks or layers. Joints and bedding planes are modelled explicitly. Roof instability is caused by the fall of roof blocks or the collapse of roof layers. The numerical codes used to analyse discontinua include distinct element codes such as UDEC and its 3D version 3DEC (Itasca, 1996, 1994) and the particle flow codes (Sakaguchi et al., 1999 and Itasca 1999).

UDEC is the most commonly used discontinuum code for span stability assessment. It requires detailed knowledge of a range of geological and geotechnical parameters. These parameters include:

joint distribution, joint strength

bedding spacing, bedding strength

intact rock strength and deformability

in situ stresses

Experience has shown that UDEC models are sensitive to the values selected for the input parameters. Consequently considerable numerical modelling experience is required to properly set up and run a model and to explain properly the results that are produced.

4.4.4 Empirical tools

A number of empirical tools have been developed to assess span stability but they do not distinguish between failure mechanisms. Most of the empirical methods are based on rock classification systems. There are three commonly used methods: Rock Mass Rating (RMR) system (Bieniawski, 1993), Q-system (Barton et al., 1974) and Coal Mining Roof Rating (CMRR) system (Molinda and Mark, 1994). Both the RMR system and the Q-system were developed for hard rock conditions and are not widely used in the coal mining industry.

The CMRR system was developed by the U.S. Bureau of Mines as an engineering tool to quantify descriptive geological information for use in coal mine design and roof support selection. The CMRR weights the geotechnical factors that contribute to roof competence such as bedding spacing, jointing and ground water, and combines them into a single rating on a scale of 0 – 100.

Recent studies conducted in underground place change development headings have shown that the CMRR can be used to evaluate unsupported span stability (Mark, 1999). The results are summarised in Figure 4.7.

The equation for the “extended cut” line shown in Figure 4.7 (the upper line) is as follows:

$$CMRR_{crit} = 19.2 + 5.38W_e \quad (4.4)$$

where W_e is the entry width in metres. Equation (4.4) separates the “sometimes stable” from the “always stable” groups. For a highwall mining entry width of 3.5 m (as fixed by the CHM cutting head), equation (4.4) implies $CMRR_{crit} = 38$.

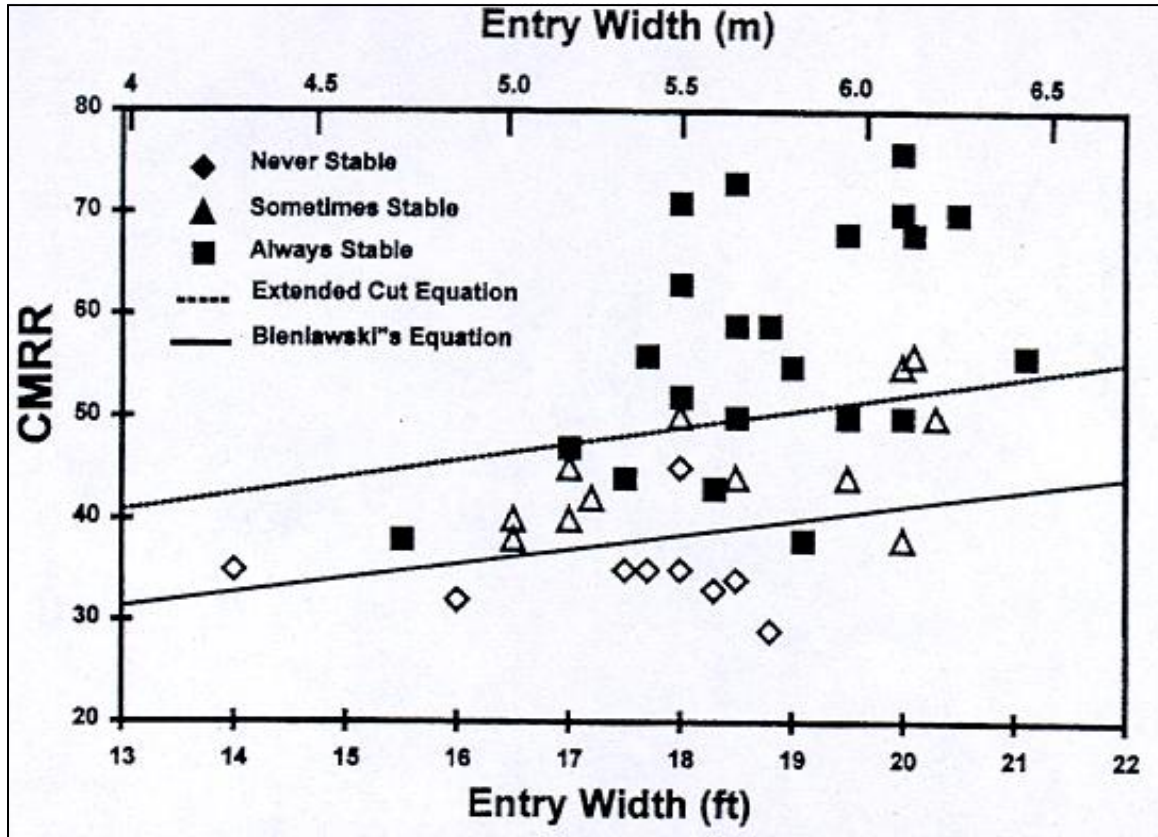


Figure 4.7. Relationship between CMRR and entry width for unsupported spans (after Mark, 1999).

Equation (4.4) is based on data from roadway spans in underground coal mines. In a highwall mining entry, the overburden stress is much lower and the unsupported entry is much longer than that of the “extended cut” (which is often no more than 6-12 m) in an underground roadway. There is very limited data where the CMRR system has been applied in highwall mining span stability assessment. The only reported application to highwall mining was in a mine in Central Queensland. The CMRR system was used to assess the span stability of the 3.5 m roadway, and the prediction results did not agree with the actual span performance possibly due to the lack of accurate geological information. Accordingly, using the CMRR system in highwall mining should be done cautiously.

4.5 Case studies

For a given site, the most likely span failure mechanism(s) should be initially identified. Prediction tools suitable for the particular failure mechanism(s) are then selected. Depending on the type of tool selected, a range of geological and geotechnical data are needed for the analysis. These data should be obtained from the site investigation and laboratory testing. Once all the necessary input parameters are known, the assessment can be carried out. Three case studies are presented which demonstrate how the prediction tools can be applied to the highwall mining span stability assessment. The procedure adopted in each case was to identify the most likely span failure mechanism(s), select the appropriate tools to analyse the mechanisms and then to analyse the mechanisms using the available geological and geotechnical data.

Case study 1

Highwall mining was to be carried out in a pit in Central Queensland. The whole seam thickness was to be cut during mining so that the immediate roof rock would be exposed in the entry spans.

The immediate roof (<5 m) was mainly composed of laminated siltstone. The thickness of the siltstone laminae ranged from 20 mm to 300 mm. The roof rock was weak (UCS=10-40 MPa) and the roof bedding was slightly angled (<15°) across the entry spans. The overburden depth ranged from 60-80 metres. Joints were -widely spaced in the roof rock.

It was assessed that the most likely span failure mechanism would be delamination and snap-through. Therefore, the analytical snap-through model (LSFM) was used to assess the span stability.

Based on the information from a site investigation and from laboratory test data, the input parameters for LSFM and their probabilities over the area of highwall mining reserve were estimated and are listed in Table 4.2.

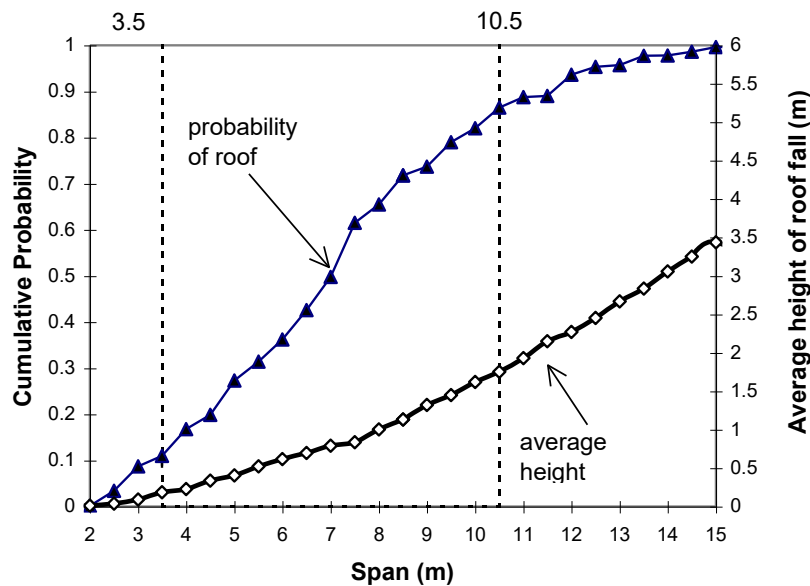
Table 4.2. Mechanical parameters of roof strata in an example highwall mining pit.

	Lowest	Low	Inter-mediate	High	Highest
E (MPa) Probability	5000 0.25		10000 0.5		20000 0.25
t (m) Probability	0.025 0.1	0.05 0.2	0.1 0.4	0.17 0.2	0.25 0.1
p Probability	1.0×selfweight 0.8		1.5×selfweight 0.15		2.0×selfweight 0.05
σ_c (MPa) Probability	15 0.3		25 0.5		35 0.2

Using Equation (4.1) and the probabilistic approach described in Shen and Duncan Fama (1996), the probability of roof falls and the average height of the falls were obtained for a range of span widths (2-15 m), and are shown in Figure 4.8.

For an ordinary highwall mining entry with a span width of 3.5 m, the model predicted an 11% chance of roof falls. The average height of the falls is predicted to be 0.2 m. The chance of roof falls increases with increasing span width, as does the height of the roof falls. For instance, with a 10.5 m span, the probability of a roof fall is 86% and the average height of fall is 1.8 m. This extent of roof fall is considered to be severe, and would likely damage and perhaps, bury the mining equipment.

The actual mining records from this pit show that some minor roof falls did occur during mining, mainly as thin roof slabs. On one occasion, a 10-11 m span was created by the convergence of three entries owing to poor mining guidance. A roof fall extending to a height



of about 2 m occurred. The observed span failure agrees well with the prediction.

Figure 4.8. Predicted probability of roof falls and the average height of the falls using the Laminated Span Failure Model (LSFM), case study 1.

Case study 2

A highwall mining pit in Central Queensland had a 4.8 m thick coal seam. Due to the limited cutting height capacity of the continuous miner to be used, a layer of coal, approximately 1.0 m thick, was to be left in the roof. The stability of the roof coal was a concern, as one metre thick coal falls can disrupt mining operations.

The Coal Roof Failure Model (CRFM) was used to assess the stability of the coal layer.

The following mechanical and geometrical parameters were selected for input to the model. These values were based on the results of a site investigation of the pit:

Span width (s)	3.5 m
Pillar width (s_p)	3.7 m
Roof coal thickness (t)	1.0 m
Young's modulus of roof rock (E_r)	3 GPa
Poisson's ratio of roof rock (ν_r)	0.25
Young's modulus of coal (E_c)	2 GPa
Poisson's ratio of coal (ν_c)	0.23
Friction angle of coal cleats (ϕ)	41°
Cohesion of coal cleats (c)	1.1 MPa
Friction angle of roof/coal interface (ϕ_i)	16°

The estimated overburden stress ranged up to 2.7 MPa over a penetration depth of 0-300 m. for the analysis, the maximum overburden stress of 2.7 MPa was used. The ratio of horizontal stress to overburden stress in the coal seam was not known but it was considered that it could be as low as $\sigma_H/\sigma_v = 0.5$ based on limited stress measurement results in adjacent areas. Three cases of low to high stress ratio ($\sigma_H/\sigma_v = 0.5, 0.75, 1.0$) were investigated.

Using the CRFM model and Equation (4.2), the calculated Factors of Safety for the three stress ratios were:

σ_H/σ_v	FoS
0.50	0.83
0.75	1.02
1.0	1.21

The results indicated that if the stress ratio in this pit was 0.5, then the Factor of Safety of the coal roof would be well below 1.0 and unstable coal roof would almost certainly result. In contrast, if the stress ratio was 1.0, the Factor of Safety of the coal roof would be about 1.2 and this would likely result in a stable coal roof. No measurements of in situ stress had been carried out in this pit so no more definitive predictions could be made.

During the actual mining of this pit, extensive roof falls were experienced from which it was inferred that low in situ horizontal stresses existed in the pit.

Case study 3

A pit in Central Queensland was to be highwall mined. The coal seam thickness was 2.1 m. Very closely spaced angled coal joints were found in the coal seam. The coal seam was overlain by a layer of 0.8 m thick, weak, laminated mudstone. It was expected that the mudstone layer would fall and cause problems during mining. It was thought that leaving coal in the roof might improve the roof conditions. An assessment was required as to how much coal should be left in the roof to ensure the span stability.

This case involved the stability of both jointed roof coal and overlying laminated mudstone. Consequently the numerical code UDEC was selected to analyse the case owing to its flexibility in handling complex geology.

UDEC models require comprehensive geological and geotechnical data. The mechanical properties relevant to the pit studied are listed in Table 4.3. The values of the properties were estimated based on the results of site investigation and previous modelling experience.

One highwall mining entry and two half pillars, one at each side of the entry were modelled. The entry was excavated in one step. This model simulated the ultimate span stability when highwall entries at both sides of an entry were excavated. The bedding in the roof mudstone, sub-vertical joints in the roof and inclined joints in the coal seam were all modelled explicitly. The roof bedding thickness was modelled to vary randomly with an average thickness of 0.1 m. Two cases were studied where roof coal thickness was 0.3 m and 0.5 m respectively.

An overburden depth of 70 m was modelled. The horizontal/vertical stress ratio was assumed to be 0.5 in the coal seam and 1.0 in the roof rock.

The modelling results for the two cases with different coal thickness are plotted in Figure 4.9 and Figure 4.10. When the roof coal was 0.3 m thick in the model the results predicted that the coal layer and 0.4 m of roof mudstone would fall. When the roof coal was 0.5 m thick in the model, a stable span was predicted.

During the actual mining operations, a coal layer was left in the roof of the first two entries in the pit. The thickness of the roof coal, however, was difficult to control, and it ranged from 0.2 m to 0.5 m. Seven coal falls were reported during mining of these two entries. After that, no roof coal was left in the entries and the weak roof mudstone was cut during mining. The rock roof, dominated by stronger siltstone, was mostly stable.

Table 4.3. Mechanical properties used for UDEC modelling of span stability.

	Young's Modulus E (MPa)	Poisson's Ratio ν	Unconfined Comp. Strength UCS (MPa)	Cohesion c (MPa)	Friction angle ϕ ($^{\circ}$)	Tensile Strength σ_t (MPa)
Coal	2000	0.23	4.8	1.09	41	0.1
Roof mudstone	3000	0.25	12.5	2.8	41	0.1
Floor mudstone	2000	0.25	4.05	0.93	41	0.2
Roof siltstone	5000	0.25	(elastic)			
Sandstone	10000	0.25	(elastic)			
Coal cleat				0.5	30	0
Coal joint				0.55	41	0
Mudstone bedding				0.1	26	0
Coal/roof/floor interface				0.1	26	0
Rock joint				0	35	0
Rock/rock interface				0.5	35	0

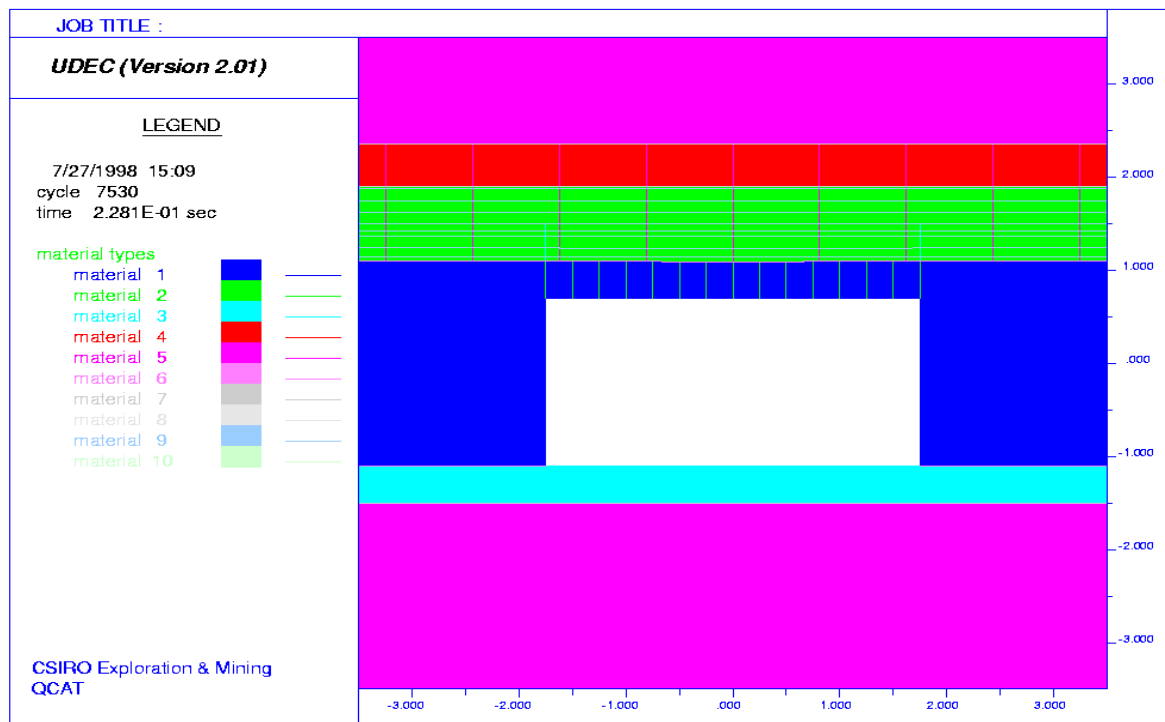


Figure 4.9. Modelling results - unstable span with 0.3 m coal roof.

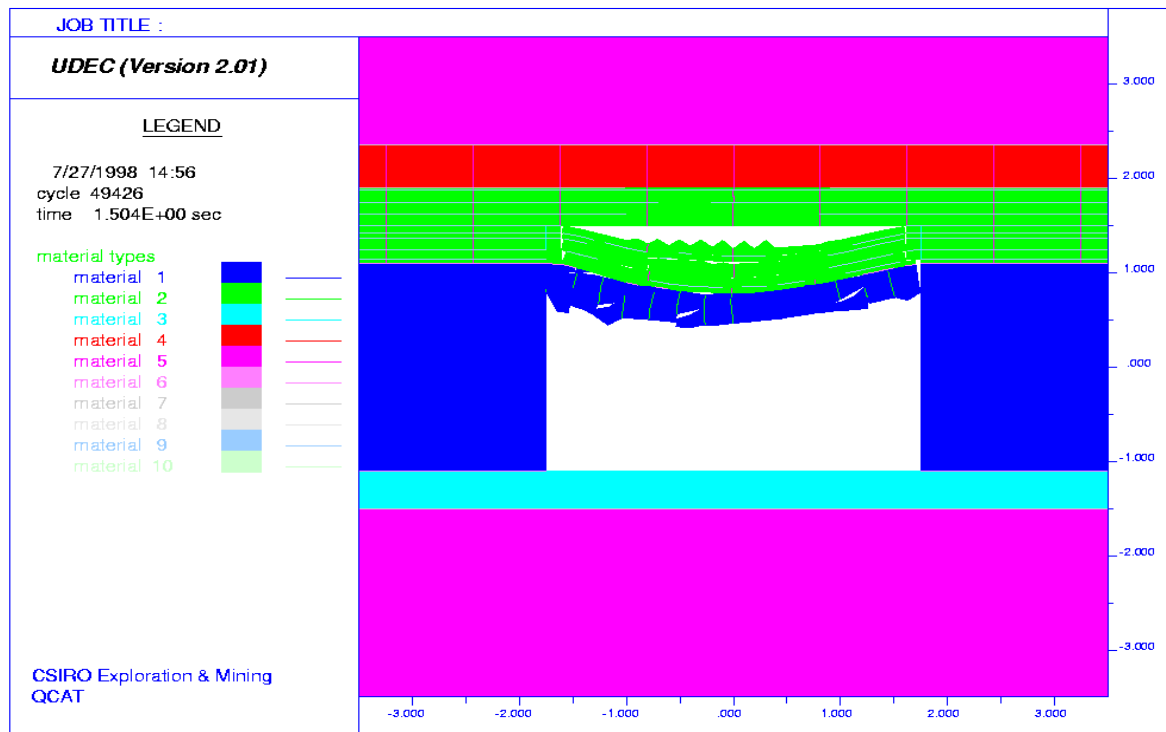


Figure 4.10. Modelling results - stable span with 0.5 m coal roof.

4.6 Conclusion

It is recommended that three steps be followed to effectively evaluate the unsupported span performance for highwall mining: They are:

Step 1: Identify the span failure mechanisms;

Step 2: Select proper tools for span stability assessment;

Step 3: Assess the stability of entry spans using all of the available geological and geotechnical data.

The most common mechanisms of roof falls in highwall mining conditions are:

- snap-through of roof layers;
- shear failure of roof coal;
- rock block falls due to jointing.

The recommended assessment tools for each or a combination of the above mechanisms are:

- Laminated Span Failure Model (LSFM) (snap-through);
- Coal Roof Failure Model (CRFM) (coal roof);

Numerical codes (e.g. UDEC or 3DEC) (block falls, and/or other mechanisms).

Coal Mine Roof Rating (CMRR) (combination of the above mechanisms)

Three application examples described in this section have demonstrated the effectiveness of the recommended procedure in highwall mining span stability assessment.

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